1. Motivation: Bottlenecks in Vanilla ViT (DINOv1)

In the standard DINOv1 baseline (`ViT-Small/16`), two core components introduce major bottlenecks:

Bottleneck 1: Absolute Position Embeddings

- **Multi-Crop Distortion**: DINO jointly trains on 224^2 (196 patches) and 96^2 (36 patches). Bicubic interpolation alters distance metrics and distorts patch aspect ratios.
- **No Translation Invariance**: Fails to directly model continuous relative displacements $(\Delta x, \Delta y)$.

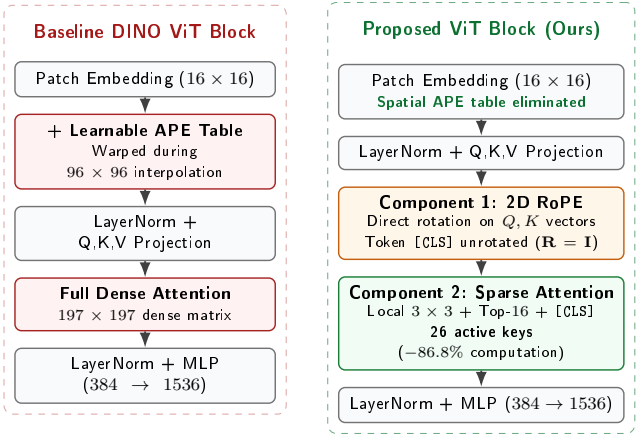
Bottleneck 2: Full Dense Attention $\mathcal{O}(N^2)$

- **Computational Redundancy**: Every token attends to all 197 tokens, yet $> 85\%$ of converged attention weights are near zero.
- **Background Clutter**: Dense attention diffuses mass into irrelevant background (sky, grass) rather than salient objects.

Our Architectural Remedy

Replace static APE with **Frequency-Adaptive 2D RoPE** & dense attention with **Spatial-Semantic Sparse Attention**.

2. Architectural Overview: Baseline vs. Proposed ViT Block



Core Design Principles

- **Strict Zero-Parameter Overhead**: Maintains exactly **21,665,664** parameters (100% match with baseline).
- **Theoretical FLOPs Parity**: **9.20 GFLOPs** at standard 224×224 input resolution.
- **Geometric & Saliency Synergy**: 2D RoPE provides continuous spatial awareness; Sparse Attention isolates the 26 most informative keys.

3. Component 1: Frequency-Adaptive 2D Rotary Position Embedding

2D Spatial Rotary Mechanism

Decompose head dimension $d_h = 64$ into spatial x, y axes:

$$d_x = d_y = 32, \quad M = 16 \text{ frequency bands } \omega_m$$

Query and key vectors rotated by orthogonal matrix $\mathbf{R}_\Theta$:

$$\tilde{\mathbf{q}}_i = \mathbf{R}_\Theta(p_{x,i}, p_{y,i})\mathbf{q}_i, \quad \tilde{\mathbf{k}}_j = \mathbf{R}_\Theta(p_{x,j}, p_{y,j})\mathbf{k}_j$$

Relative translation invariance via inner product:

$$\langle \tilde{\mathbf{q}}_i, \tilde{\mathbf{k}}_j \rangle = \mathbf{q}_i^T \mathbf{R}_\Theta(p_{x,i} - p_{x,j},\, p_{y,i} - p_{y,j})\mathbf{k}_j$$

$\Rightarrow$ Directly encodes continuous 2D relative displacement!

Frequency-Adaptive Coordinate Blending

Blends raw integer and normalized continuous coordinates:

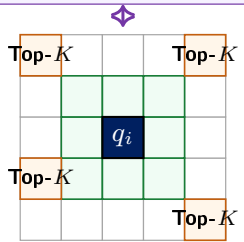
$$p_m = \alpha_m \cdot p_{\text{raw}} + (1-\alpha_m) \cdot p_{\text{norm}}, \quad \alpha_m = 1 - \frac{m}{M-1}$$

- **High Frequencies** ($\alpha \to 1$): Raw integer grid $\Rightarrow$ captures sharp local edge structures.
- **Low Frequencies** ($\alpha \to 0$): Normalized grid $\Rightarrow$ guarantees scale invariance across 224^2 and 96^2 crops.

Preserving `[CLS]` **Token**: Set $\boldsymbol{\theta}_{\text{cls}} = \mathbf{0} \Rightarrow \mathbf{R} = \mathbf{I}$, preventing artificial spatial bias on the global representation.

4. Component 2: Spatial-Semantic Sparse Softmax Attention

Bidirectional [CLS] Conduit (Index 0)
[CLS] attends to all tokens
& all tokens attend to [CLS]



Active Key Set $\mathcal{S}_i$ (26 Keys Total)

- **Local 3×3 ($\mathcal{N}_{\text{local}}$)**: 9 spatial neighbors (Chebyshev radius $R = 1$).
- **Semantic Top-16 ($\mathcal{N}_{\text{semantic}}$)**: 16 highest dot-product keys across the grid.
- **Bidirectional [CLS]**: Seamless global-to-local communication.

86.8% Computation Reduction vs. dense attention (26/197).

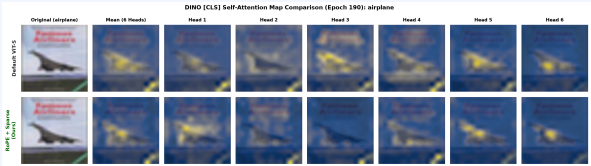
Exact Softmax Normalization

$$\mathbf{A}_{ij} = \frac{\exp(\tilde{\mathbf{q}}_i^T \tilde{\mathbf{k}}_j / \sqrt{d_h})}{\sum_{l \in \mathcal{S}_i} \exp(\tilde{\mathbf{q}}_i^T \tilde{\mathbf{k}}_l / \sqrt{d_h})} \quad (j \in \mathcal{S}_i)$$

Exact Formulation: Probabilities strictly sum to 1 over $\mathcal{S}_i$, ensuring stable gradients.

5. Visual Evidence: Attention Maps & Empirical Gains

6-Head Attention Map Comparison (Epoch 190)



Qualitative Analysis

- **Baseline ViT**: Attention is diffused across the background sky and ground.
- **RoPE + Sparse (Ours)**: Heads 2 and 5 concentrate sharply on the aircraft fuselage, wings, and tail with crisp boundary segmentation.

Quantitative Improvements (CIFAR-10)

- **k-NN Top-1 ($k = 10$)**: $74.41\% \rightarrow \mathbf{75.97}\%$ (**+1.56% gain**).
- **Linear Probing Top-1**: $81.46\% \rightarrow \mathbf{83.98}\%$ (**+2.52% gain**).
- **Early Convergence Speed**: Epoch 10: $73.76\% \rightarrow \mathbf{79.86}\%$ (**+6.10% boost**).

Takeaway for Part 2

2D RoPE + Sparse Attention eliminates geometric distortion and suppresses background noise **without adding any parameters**.